

Identification and Authentication Procedures

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System: Example System Demo (Example System Example System Demo IATT)

1. Purpose

These procedures implement NIST SP 800-53 Rev. 5 controls IA-2, IA-3, IA-4, IA-5, IA-6, IA-7, IA-8, and IA-11 on the Example System Demo system. They define how organizational and non-organizational users are uniquely identified, authenticated, and their authenticators managed.

2. Authentication Methods

User Class	Primary	Backup
Privileged (admin)	PIV smart card + PIN	FIDO2 hardware key
Standard user	Username + password + TOTP	Recovery code
Service account	Managed Service Account (gMSA)	n/a
Local console	Username + password	n/a

3. Password Requirements

Password length, complexity, age, history, and lockout settings are enforced by the Default Domain Policy GPO on the Example System Demo domain. The current export is recorded in GPO_Password_Policy_Export.xlsx (USD20240218). Minimum settings:

- Minimum length: 15 characters
- Complexity: enabled (3 of 4 character classes)
- Maximum age: 60 days
- History: 24 passwords remembered
- Lockout threshold: 5 invalid attempts in 15 minutes
- Lockout duration: 30 minutes; auto-unlock after duration

4. Authenticator Management (IA-5)

Initial authenticators are distributed out-of-band via the issuance station in Room 204. Users must change initial authenticators on first logon. Lost authenticators are reported to the help desk; identity is re-verified per Section 4.2 of USD20240315 before re-issuance.

4.1 PIV Card Lifecycle

PIV cards are issued by the Example System Demo PIV sponsor per FIPS 201-3. Card termination follows immediately on departure; issuance and termination events are recorded in the PIV management system.

5. Cryptographic Module Authentication (IA-7)

All authenticator verification uses FIPS 140-3 validated cryptographic modules. Microsoft Cryptographic Primitives Library (Bcryptprimitives.dll) certificate #4544 is in use on all Windows endpoints; YubiKey 5 FIPS series (certificate #4569) is used for FIDO2 backup.

6. References

- NIST SP 800-53 Rev. 5: IA-2, IA-3, IA-4, IA-5, IA-5(1), IA-6, IA-7, IA-8, IA-11
- FIPS 201-3 (PIV)
- FIPS 140-3 (Cryptographic Module Validation)
- NIST SP 800-63B (Digital Identity Guidelines)
- Account Management Policy (USD20240315)